Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1 Load the packages and check the working directory
library(tidyverse);library(lubridate);library(viridis);library(here);
library(rvest);library(dataRetrieval)

here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```r
#2
homework.url <-
  "https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024"

# Read the contents into a webpage object
durham.lwsp <- read_html(homework.url)
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```r
#3
System.name <- durham.lwsp %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>% html_text()
PWSID <- durham.lwsp %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>% html_text()
Ownership <- durham.lwsp %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>% html_text()
max_day_use <- durham.lwsp %>%
  html_nodes("th~ td+ td") %>% html_text()
```

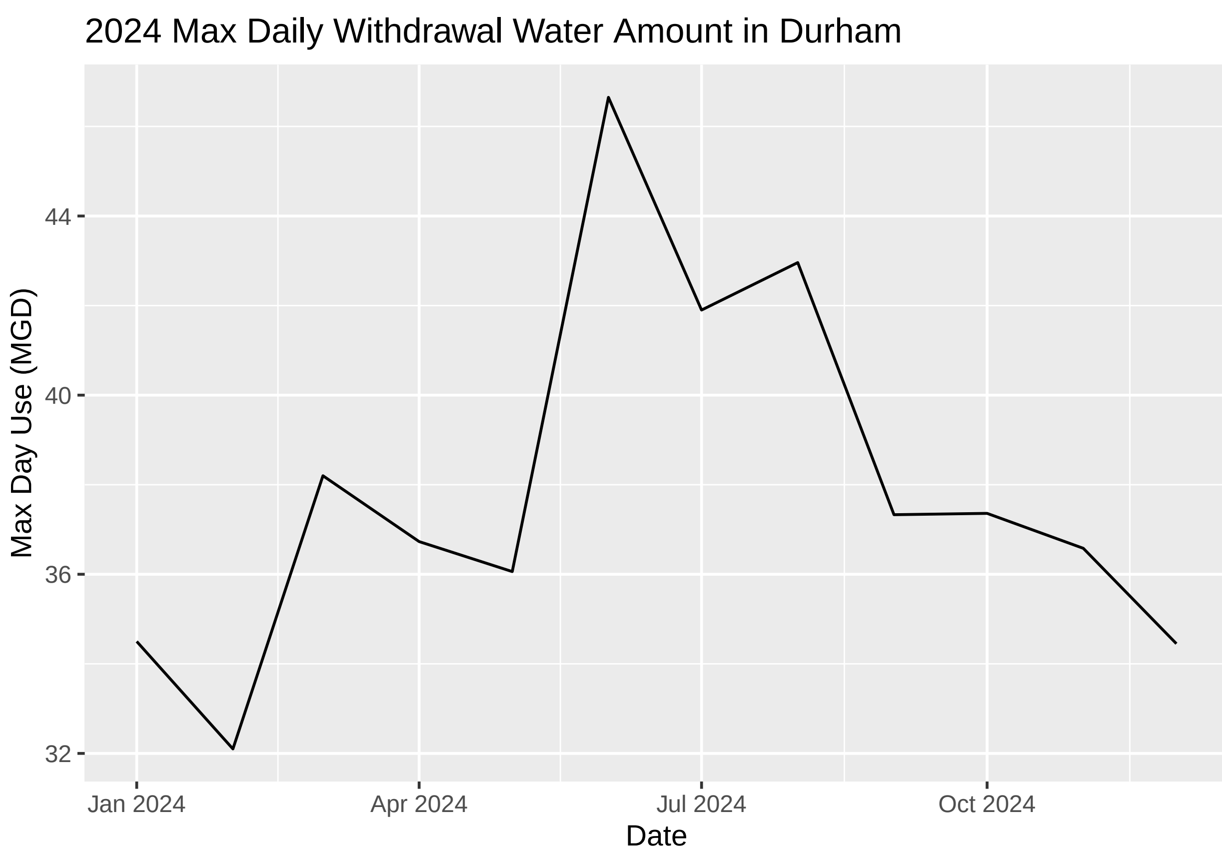
4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

   > TIP: Use `rep()` to repeat a value when creating a dataframe.

   > NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc…

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```r
#4 Create a daraframe
df_dayuse <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                             "Year" = rep(2024, 12),
                             "Max_Day_Use_mgd" = as.numeric(max_day_use))

df_dayuse <- df_dayuse %>%
  mutate(System_name = !!System.name,
         PWSID = !!PWSID,
         Date = my(paste(Month,"-",Year)))

#5 Create a line plot
ggplot(df_dayuse, aes(x = Date, y = Max_Day_Use_mgd)) +
  geom_line() +
  labs(title = paste("2024 Max Daily Withdrawal Water Amount in", System.name),
       y = "Max Day Use (MGD)", x = "Date")
```



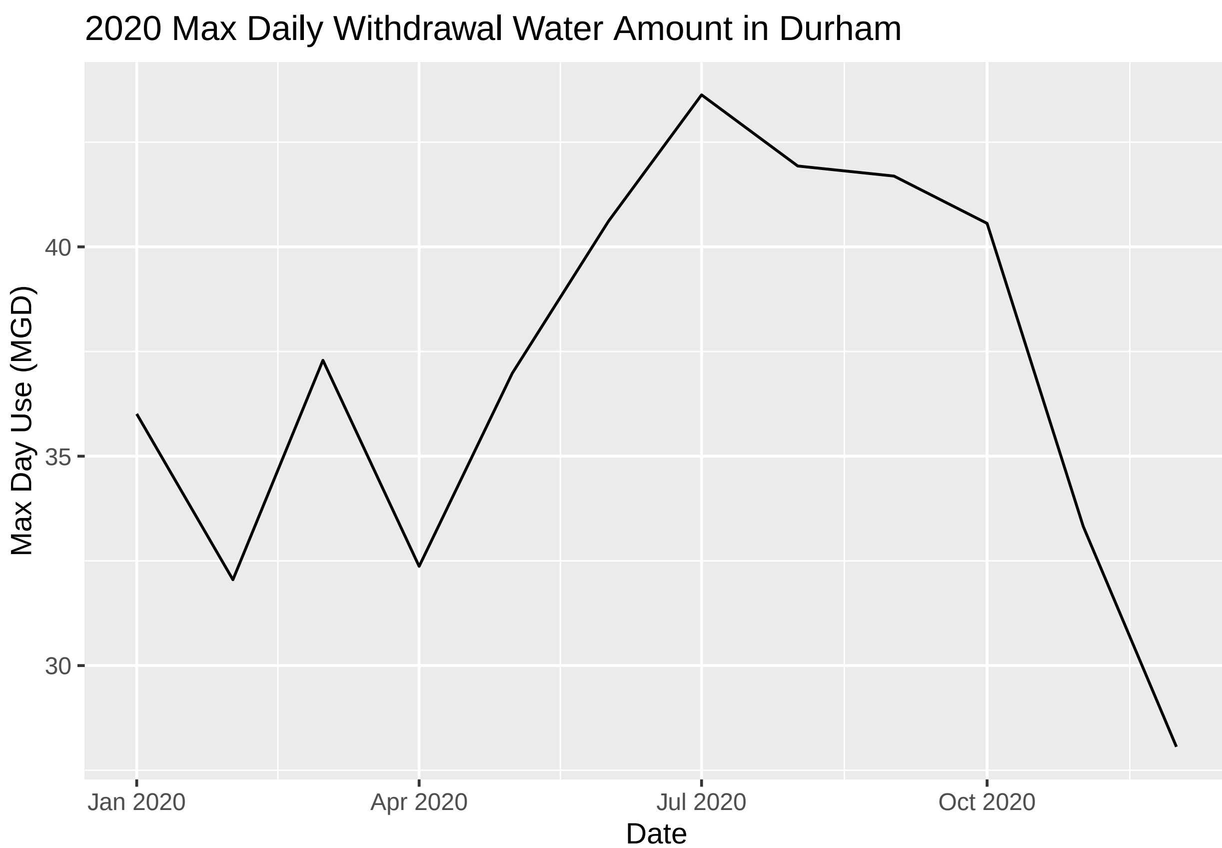
6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - "PWSID" and "year" - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function's output

```r
#6.
# Construct the function
scrape.it <- function(the_pwsid, the_year){
  the.site <- read_html(paste0
                        ("https://www.ncwater.org/WUDC/app/LWSP/report.php?",
                                "pwsid=", the_pwsid,
                                "&year=", the_year))

  # Scrape data items
  System.name <- the.site %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>% html_text()
  Ownership <- the.site %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>% html_text()
  max_day_use <- the.site %>%
  html_nodes("th~ td+ td") %>% html_text()

  # Data frame
  df_dayuse <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                              "Year" = rep(the_year, 12),
                              "Max_Day_Use_mgd" = as.numeric(max_day_use))

  df_dayuse <- df_dayuse %>%
  mutate(System_name = !!System.name,
         PWSID = !!PWSID,
         Date = my(paste(Month,"-",Year)))

  # Line plot
  ggplot(df_dayuse, aes(x = Date, y = Max_Day_Use_mgd)) +
  geom_line() +
  labs(title = paste(the_year, "Max Daily Withdrawal Water Amount in",
                     System.name), y = "Max Day Use (MGD)", x = "Date")

  # Return the dataframe
  return(df_dayuse)

}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```r
#7
Durham.2020.lwsp <- scrape.it("03-32-010", 2020)

# Line plot
ggplot(Durham.2020.lwsp, aes(x = Date, y = Max_Day_Use_mgd)) +
  geom_line() +
  labs(title = paste("2020 Max Daily Withdrawal Water Amount in Durham"),
       y="Max Day Use (MGD)", x="Date")
```

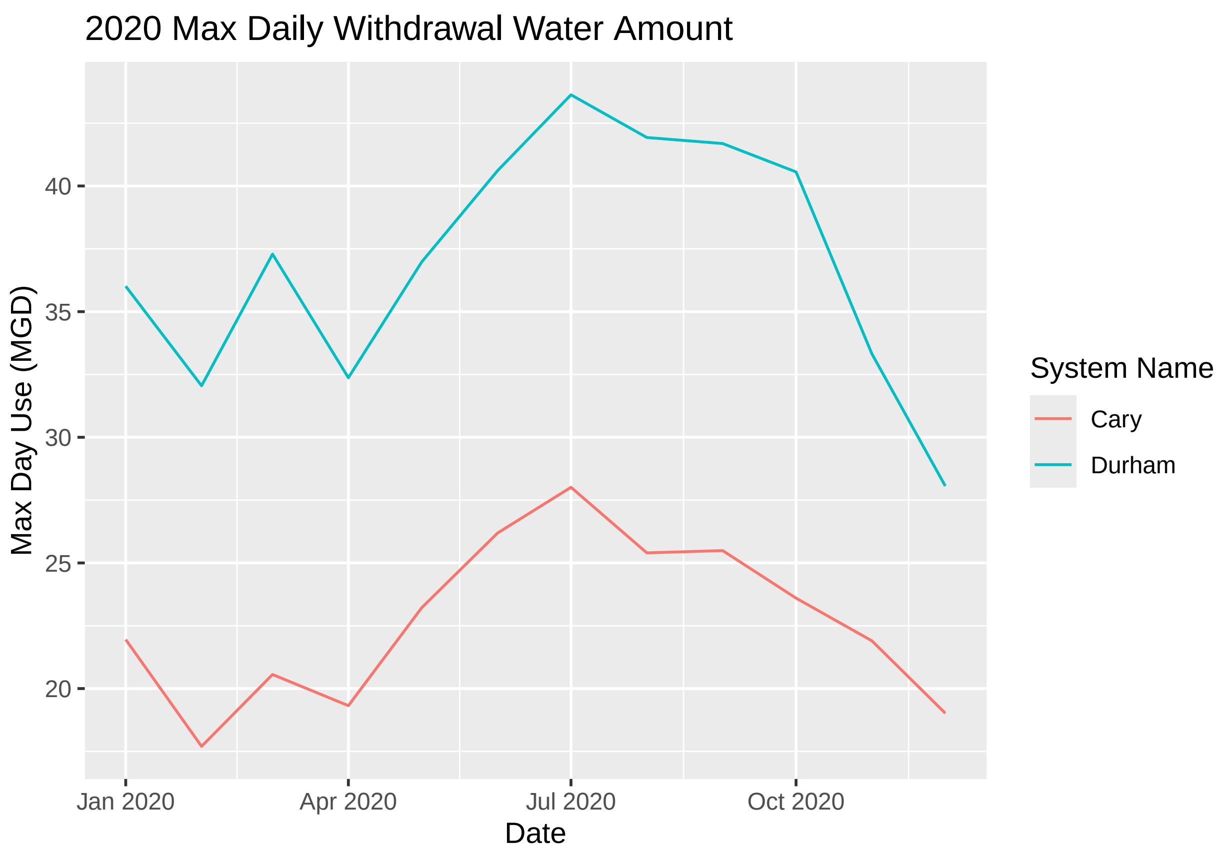



```r
str(Durham.2020.lwsp)
```

```
## 'data.frame':    12 obs. of  6 variables:
##  $ Month          : num  1 5 9 2 6 10 3 7 11 4 ...
##  $ Year           : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ Max_Day_Use_mgd: num  36 37 41.7 32 40.6 ...
##  $ System_name    : chr  "Durham" "Durham" "Durham" "Durham" ...
##  $ PWSID          : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
##  $ Date           : Date, format: "2020-01-01" "2020-05-01" ...
```

8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```r
#8
Cary.2020.lwsp <- scrape.it("03-92-020", 2020)

lwsp.combined <- rbind(Durham.2020.lwsp, Cary.2020.lwsp)

# Line plots combined
ggplot(lwsp.combined, aes(x = Date, y = Max_Day_Use_mgd, color = System_name)) +
  geom_line() +
  labs(title = paste("2020 Max Daily Withdrawal Water Amount"),
       y = "Max Day Use (MGD)", x = "Date", color = "System Name")
```




```
str(Durham.2020.lwsp)
```

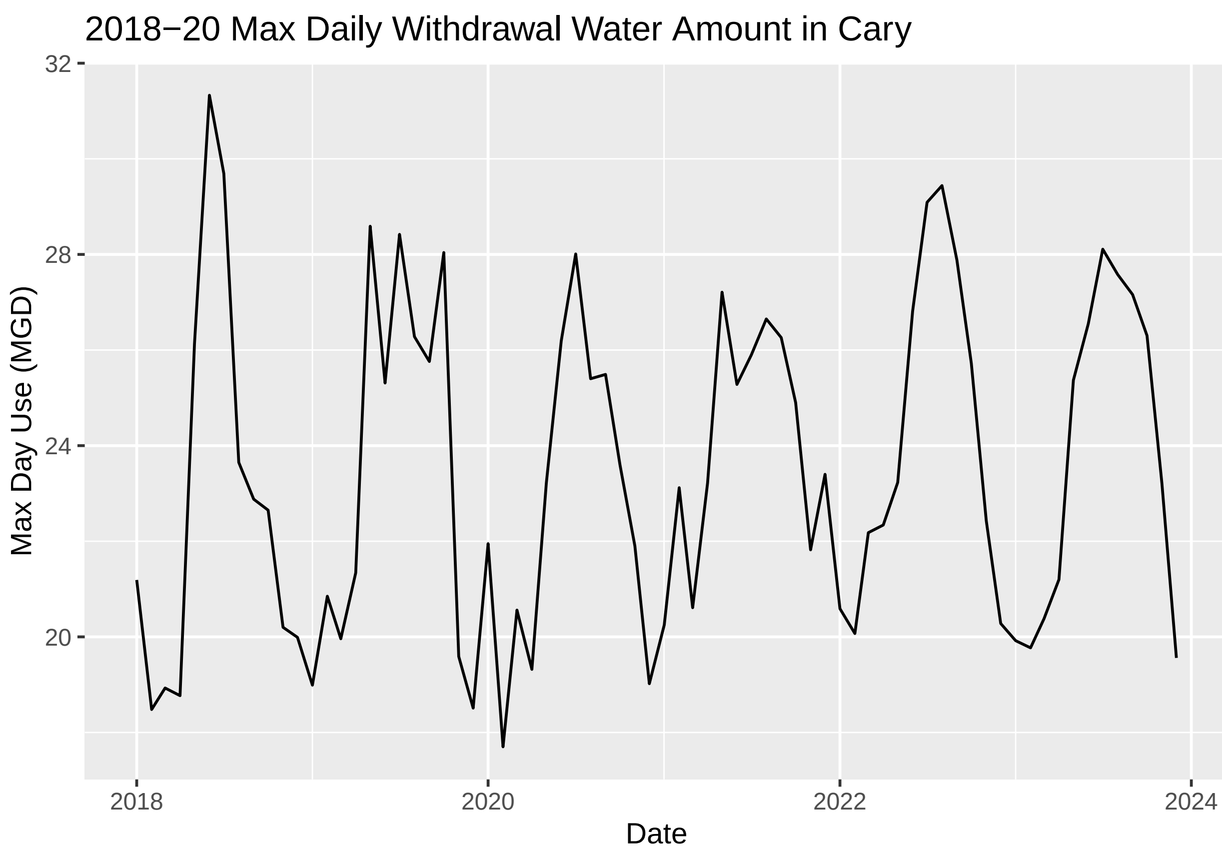
```
## 'data.frame':    12 obs. of  6 variables:
##  $ Month          : num  1 5 9 2 6 10 3 7 11 4 ...
##  $ Year           : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
##  $ Max_Day_Use_mgd: num  36 37 41.7 32 40.6 ...
##  $ System_name    : chr  "Durham" "Durham" "Durham" "Durham" ...
##  $ PWSID          : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
##  $ Date           : Date, format: "2020-01-01" "2020-05-01" ...
```

9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023.Add a smoothed line to the plot (method = 'loess').

   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```
#9
the_years = rep(2018:2023)
the_system = "03-92-020"

# Use lapply to apply the scrape function
cary.20182020.dfs <- lapply(X = the_years,
                  FUN = scrape.it,
```

```r
                   the_pwsid = the_system)

# Can also use map2 to run the function iteratively
the_system.map2 <- rep.int("03-92-020", length(the_years))
cary.20182020.dfs.map2 <- map2(the_system.map2, the_years, scrape.it)

cary.20182020.df <- bind_rows(cary.20182020.dfs)

ggplot(cary.20182020.df, aes(x = Date, y = Max_Day_Use_mgd)) +
  geom_line() +
  labs(title = paste("2018-20 Max Daily Withdrawal Water Amount in Cary"),
       y = "Max Day Use (MGD)", x = "Date")
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time?

> Answer: In general, the water usage changed on an annual basis. It rose in early summer and dropped in fall. Though there were some fluctuations, overall the trend was quite stable.